



# Parul University

FACULTY OF ENGINEERING AND TECHNOLOGY

Parul Institute of Engineering Technology

**Cyber Security Laboratory**

**(303105342)**

6<sup>th</sup> SEMESTER (3<sup>RD</sup> YEAR)

COMPUTER SCIENCE & ENGINEERING DEPARTMENT

# LAB MANUAL



## CERTIFICATE

This is to certify that Mr. **KUNAM MANI SAI CHARAN REDDY** with enrollment No **2303031250067** has successfully completed his/her laboratory experiments in the **Cyber Security Laboratory** (303105342) from the department of COMPUTER SCIENCE & ENGINEERING during the academic year 2025-2026.

Date of Submission: .....

Staff In charge: .....

Head of Department: .....

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## **PRACTICAL NO: 1**

**Aim:** Implementation to gather information from any PC's connected to the LAN using who.is, port scanners, network scanning, Angry IP scanners etc.

**Objective :** To know how to gather information about the networks by using different n/w reconnaissance tools.

**Requirements :** Laptop, who.is, nmap, angry ip scanner.

### **1. Who.is**

- **Theory**

whois search for an object in a WHOIS database. WHOIS is a query and response protocol that is widely used for querying databases that store the registered users of an Internet resource, such as a domain name or an IP address block but is also used for a wider range of other information. Most modern versions of whois try to guess the right server to ask for the specified object. If no guess can be made, whois will connect to whois.networksolutions.com for NIC handles or whois.arin.net for IPv4 addresses and network names.

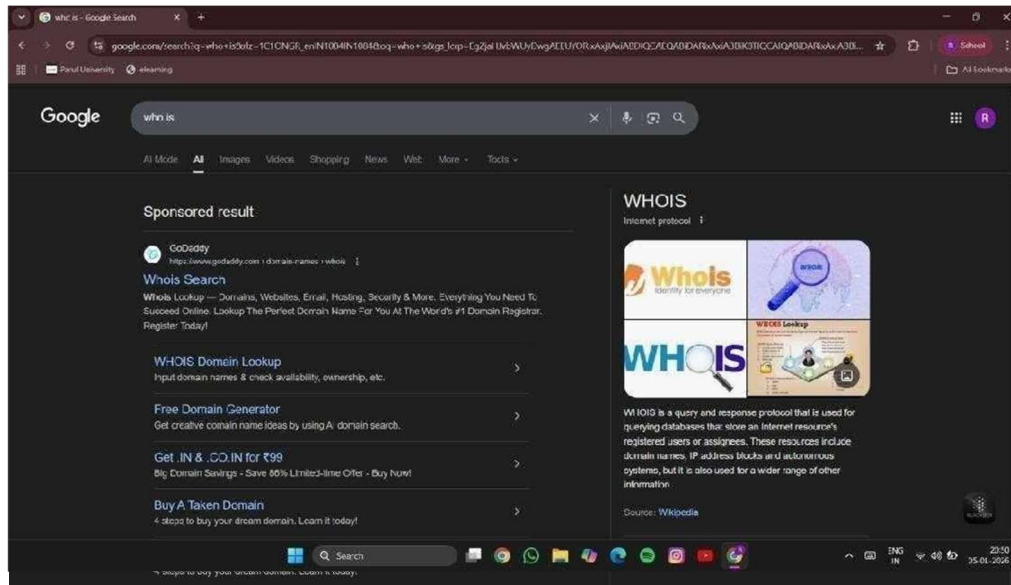
#### **EXAMPLES:**

- Obtaining the domain WHOIS record by IP Querying
- WHOIS record by IP querying
- Querying WHOIS in google search engine

To use the WHO.IS lookup tool, just enter the domain name whose information you'd like to view into the search field on the WHOIS main page. You can retrieve key data about a domain in this way, including availability, **domain owner lookup**, and creation and expiration details. If you own multiple domains of your own, it can be helpful to download exportable lists from the tool to analyze large amounts of domains data.

- Implementation

Step-1: search who.is in web browser



Step-2: open the who.is tool

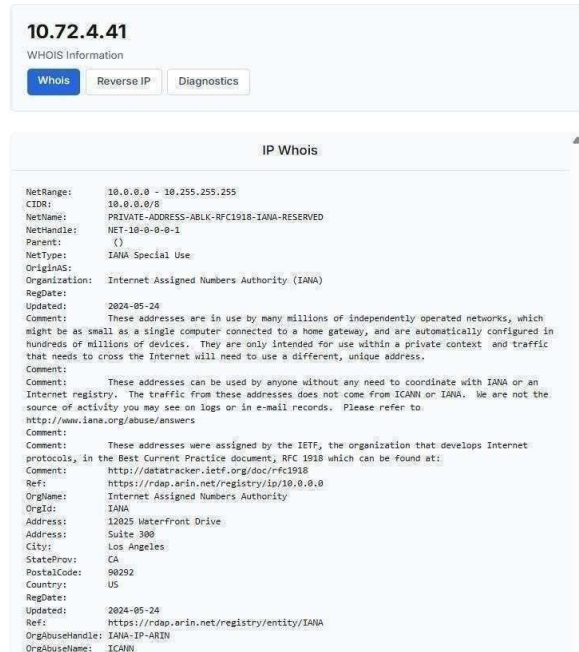


Step-3: Enter the Domain name or IP Address of which devices' information you want to gather



- **Observation**

Get the information about searched domain name or IP Address



## 2. Port Scanner

Nmap is convenient during penetration testing of networked systems. Nmap provides the network details, and also helps to determine the security flaws present in the system. Nmap is **platform-independent** and runs on popular **operating systems** such as **Linux**, **Windows** and **MAC**

Nmap is a useful tool for network scanning and auditing purposes.

- It can search for hosts connected to the Network.
- It can search for free **ports** on the target host.
- It detects all services running on the host with the help of **operating system**.
- It also detects any **flaws** or **potential vulnerabilities** in networked systems.

It is effortless to work with the Nmap. With the release of a new graphical user interface called **GenMap User**, it performs many tasks such as saving and comparing scan results, scanning the results in a database, and visualize the network system topology graphically, etc.

- **Advantage of Nmap:**

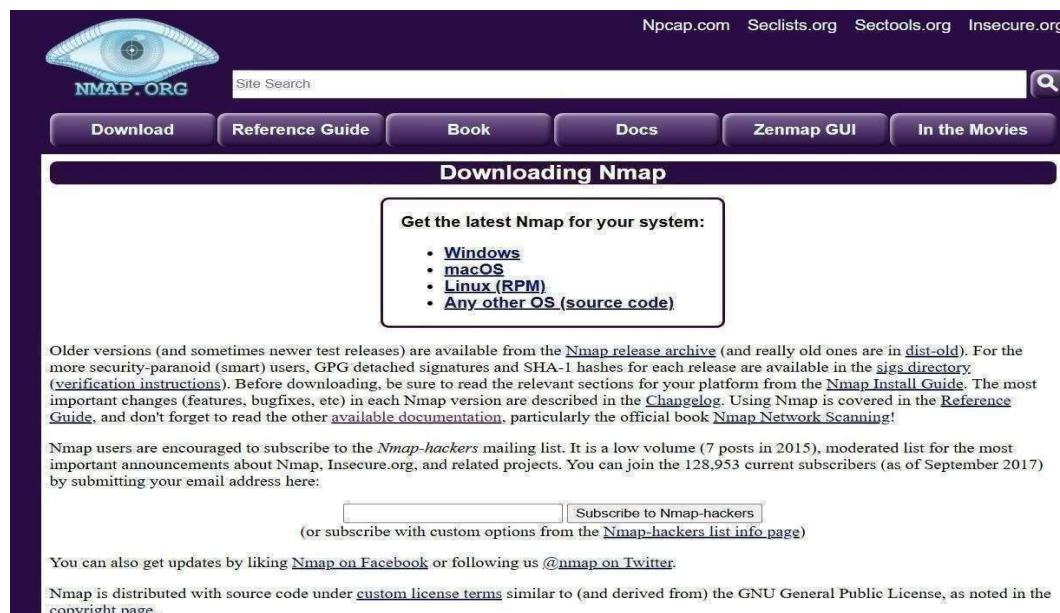
Nmap has a lot of advantages that make it different from open-source and **free** to use.

Some other advantages are listed below.

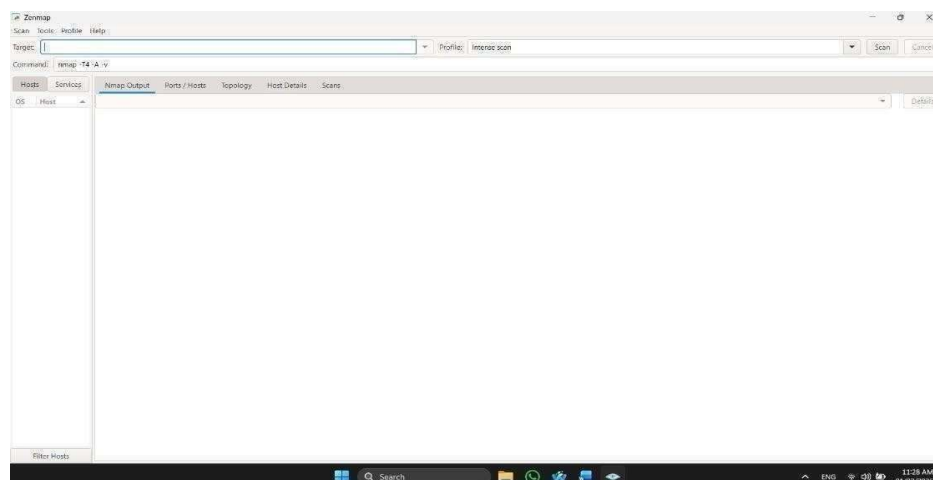
- It is used for auditing network systems as it can detect new servers.
- It will search for **subdomain** and Domain Name System.
- With the help of Nmap Scripting Engine (**NSE**), interaction can be made with the target host.
- It determines the nature of the service in the host and performs whether the host is a mail service or a web server

## • Implementation

Step-1: Install Nmap in Windows



Step-2: Open Nmap





Step-3: Open command prompt and give command ipconfig to see local network

```

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

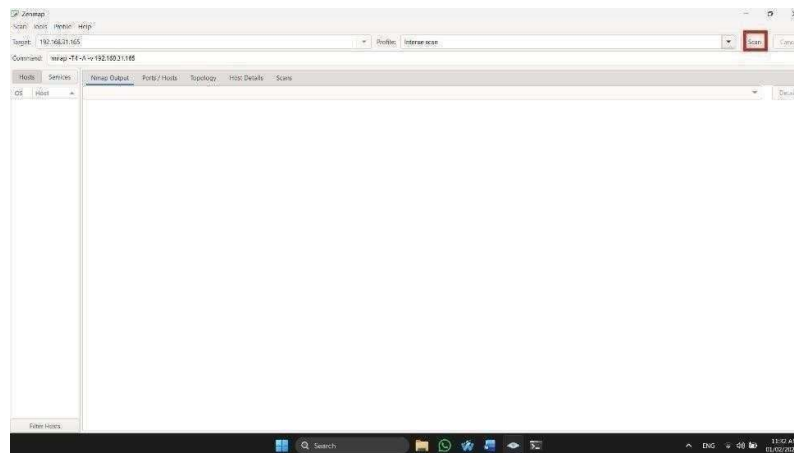
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::6a59:4d60:311f:14be%14
    IPv4 Address. . . . . : 10.72.4.41
    Subnet Mask . . . . . : 255.255.224.0
    Default Gateway . . . . . : 10.72.0.1

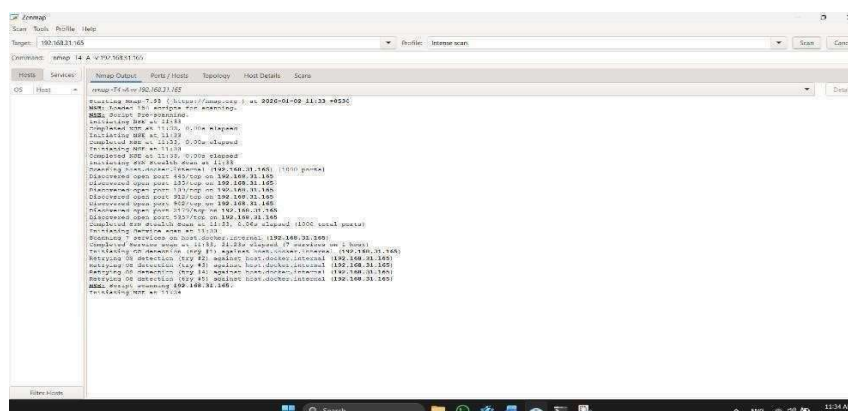
Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
  
```

Step 4: paste ip address in target select scan type and press scan button



**Nmap Output:**



Step-5: Open Command prompt Run as administrator  
To see the open ports run:

```
C:\Windows\System32>nmap solvetic.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-02 19:01 +0530
Nmap scan report for solvetic.com (178.33.118.246)
Host is up (0.22s latency).
rDNS record for 178.33.118.246: mail.solvetic.com
Not shown: 984 closed tcp ports (reset)
PORT      STATE      SERVICE
21/tcp    open       ftp
22/tcp    open       ssh
25/tcp    filtered   smtp
53/tcp    open       domain
80/tcp    open       http
110/tcp   open       pop3
135/tcp   filtered   msrpc
139/tcp   filtered   netbios-ssn
143/tcp   open       imap
443/tcp   open       https
445/tcp   filtered   microsoft-ds
465/tcp   open       smtps
587/tcp   open       submission
993/tcp   open       imaps
995/tcp   open       pop3s
2875/tcp  filtered   dxmessagebase2

Nmap done: 1 IP address (1 host up) scanned in 92.08 seconds
```

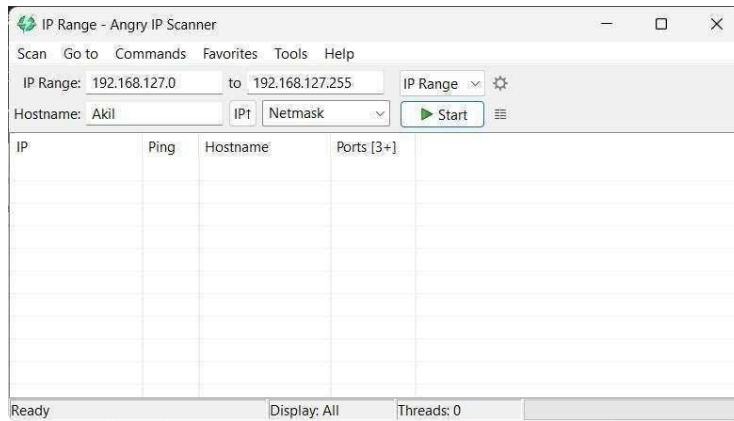
### 3. Angry IP Scanner

- **Implementation**

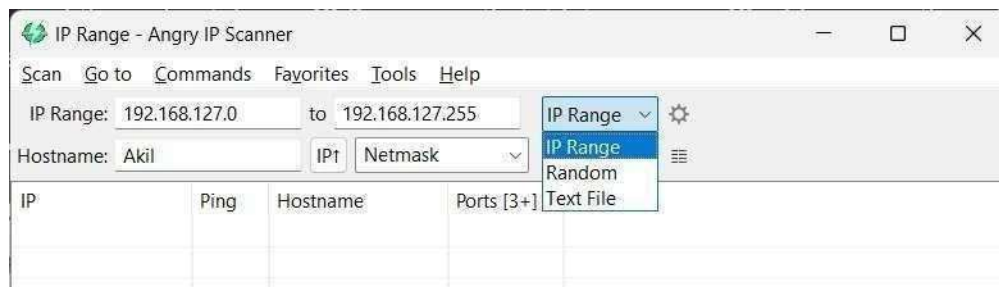
Install Angry IP Scanner



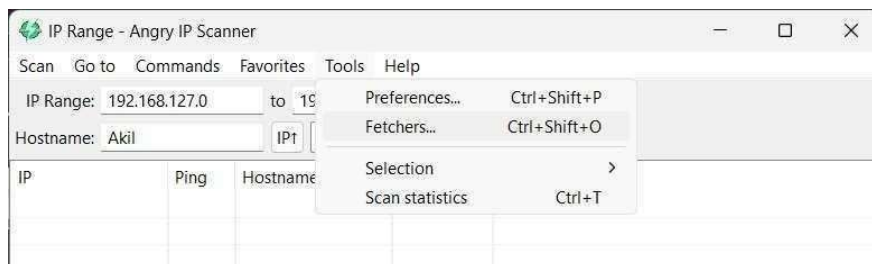
Once installed, open the application by searching for it in the Start Menu. As you can see, the home screen of the application is pretty simple and straightforward. By default, Angry IP scanner will enter your local IP address range and your computer name as the hostname.



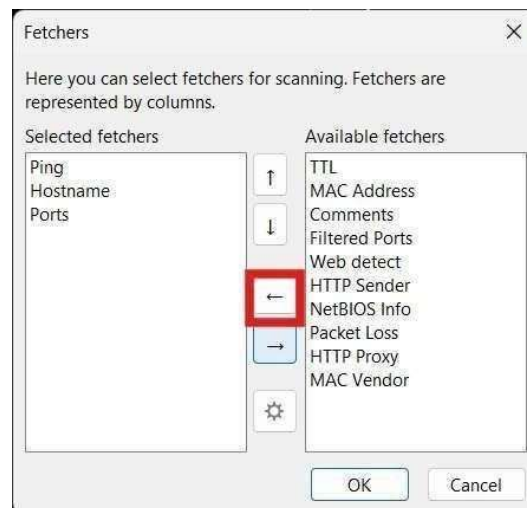
The good thing about Angry IP Scanner is that it lets you scan IP addresses in three different ways. They are, the range you specified, a random IP address or a list of IP addresses from a text file. You can easily select the scan mode from the drop-down menu next to the IP address field.



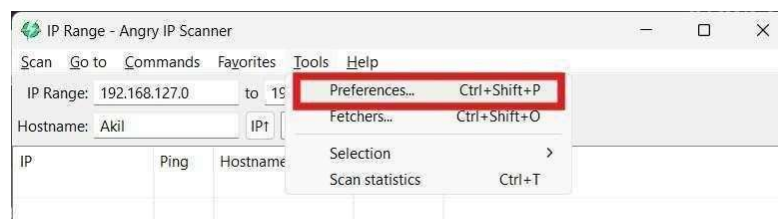
As you can see from the above image, the Angry IP Scanner will only include default fetchers like Ping, Hostname, and Ports. However, you can add more fetchers to get and see more information about an IP address. To do that, select “Tools > Fetchers.”



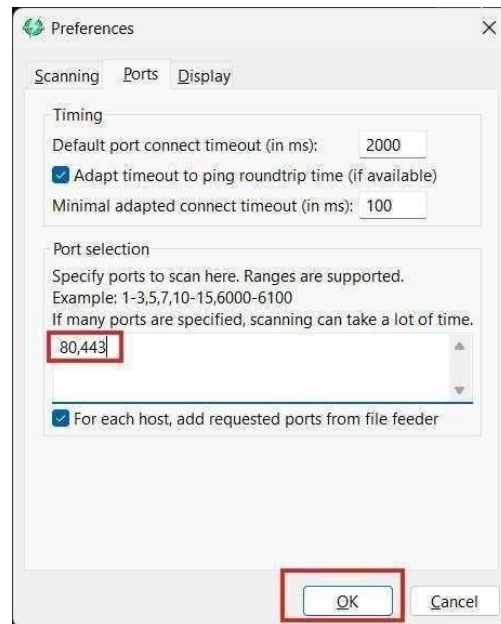
In this window, you will see all the current fetchers on the left pane and all the available fetchers in the right pane. To add a fetcher, select the fetcher on the right pane and then click on the button that looks like “Less than” sign. In my case, I’ve added new fetchers like MAC address, NetBIOS info, filtered ports, and Web detectors.



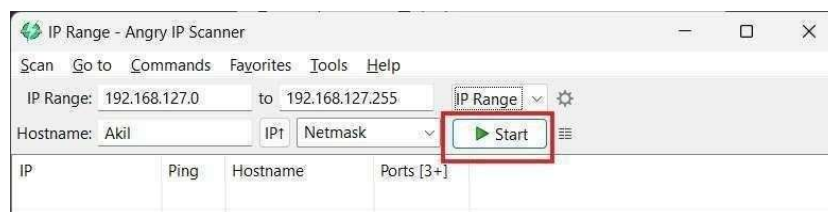
Moreover, Angry IP Scanner will only tell whether the ports are open or not. It will not list the individual ports that are open. So, if you want to do a port scan, then you need to configure the application. To do that, simply navigate to “Tools” and then select the option “Preferences.”



Here, navigate to the “Ports” tab and enter the ports you want to scan in the “Ports Selection” field. As you can see, I’m scanning for port 80 and port 443. If you want to scan a range of ports, then enter the port range like “1-1000.” After you have entered the port number, click OK to save the changes.



Once you are done configuring the Angry IP Scanner, you can continue to scan. To start off, set the scan mode to “IP Range,” enter the IP address range in the “IP address” fields and then click on the button “Start.” For instance, I’ve entered an IP range that is known to have live devices connected to it.



Depending on the number of addresses in the range, it may take some time to complete. Once completed, the application will show you a summary of the scan. The summary includes the number of hosts that are alive and the number of hosts that have open ports. Just click on the button “Close” to continue.



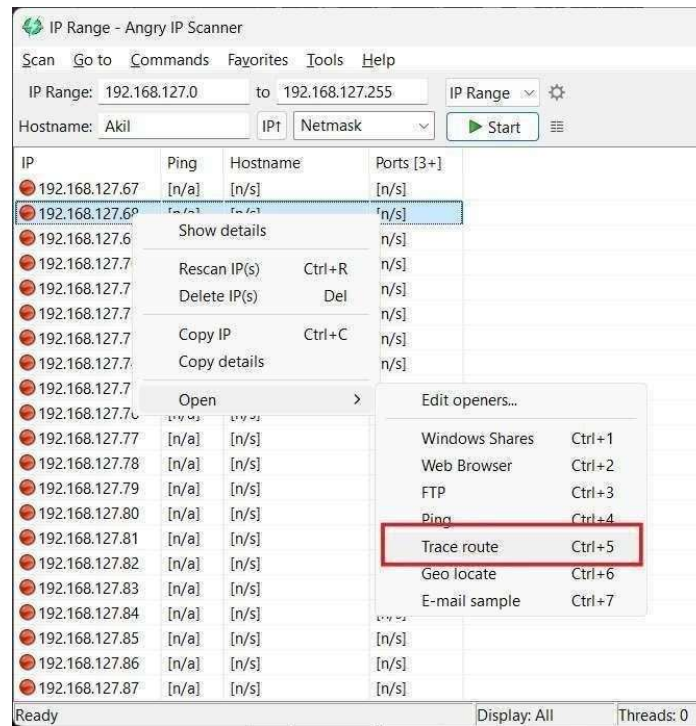
Once you close the summary window, you will see the list of all the IP addresses. You can also see additional details in different “fetcher” columns. In case you are wondering, here’s what the coloured dots next to each IP address mean.

**Red:** The IP address is inactive, dead or there is no device connected to this IP address.

**Blue:** The IP address is either active or busy and not responding to the requests sent by Angry IP Scanner. This usually will be your own IP Address.

**Green:** The IP address is active, and the device connected to it is responding to the requests made by Angry IP Scanner. There may also be open ports.

Apart from copying the details of an IP address, you can also perform a range of different activities on the entries. You can open an IP address in the web browser, do an FTP, trace routing, etc. For instance, if you want to traceroute an IP address, simply right-click on the target IP address. After that, select the option Open and click on Traceroute.



Once you are done scanning an IP address or the IP address range, you can save the scan results. To do that, select the option Scan from the menu bar.

## PRACTICAL NO:2

**Aim:** Experiments with open source firewall/proxy packages like iptables, squid etc.

- **IP Tables:**

- Simply put, iptables is a firewall program for Linux. It will monitor traffic from and to your server using tables. These tables contain sets of rules, called **chains**, that will filter incoming and outgoing data packets.
- When a packet matches a rule, it is given a target, which can be another chain or one of these special values:
- **ACCEPT** – will allow the packet to pass through.
- **DROP** – will not let the packet pass through.
- **RETURN** – stops the packet from traversing through a chain and tell it to go back to the previous chain.

- **Iptables chains:**

- In this iptables practical, we are going to work with one of the default tables, called filter. It consists of three chains:
- **INPUT** – controls incoming packets to the server.
- **FORWARD** – filters incoming packets that will be forwarded somewhere else.
- **OUTPUT** – filter packets that are going out from your server.

- **How to Install and Use Iptables Linux Firewall**

- Installing of Iptables
- sudo apt-get update
- sudo apt-get install iptables
- Check the status of your current iptables configuration by running.
- sudo iptables -L

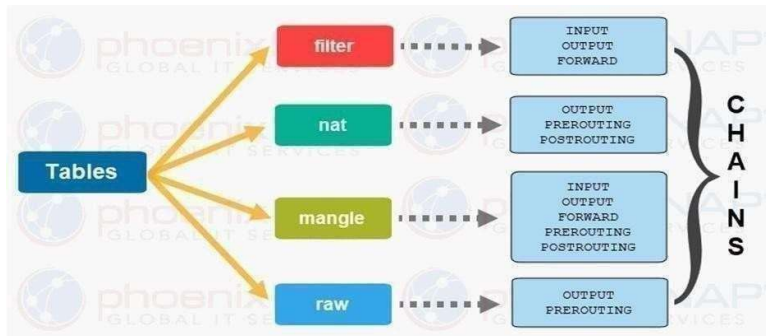
```
(kali@kali)-[~]
$ sudo iptables -L
[sudo] password for kali:
Chain INPUT (policy ACCEPT)
target     prot opt source                destination

Chain FORWARD (policy ACCEPT)
target     prot opt source                destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source                destination
```



- Types of tables in iptables



- Commands:

- Iptables -A INPUT -s your ip -j DROP
- Iptables -A INPUT -s your ip -j ACCEPT
- ➔ to stop https websites
- ➔ to stop http websites
- Iptables -A OUTPUT -p tcp -o eth0 -s your ipaddress --dport 80 -j DROP

- Follow the instructions to create a firewall rule that prevents the communication between two systems via ping.

Step-1 : CHECK YOUR IPADDRESS IN BOTH OPERATING SYSTEM(**KALI LINUX AND WINDOWS**) BY FOLLWING COMMANDS IN THE BELOW PICTURES.

```
(kali@kali)~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.236.128 netmask 255.255.255.0 broadcast 192.168.236.255
    inet6 fe80::d96a:4096:b6e2:2076 prefixlen 64 scopeid 0<link>
    ether 00:0c:29:4d:b2:ec txqueuelen 1000 (Ethernet)
    RX packets 27 bytes 12184 (11.8 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 55 bytes 12436 (12.1 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 8 bytes 480 (480.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 8 bytes 480 (480.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
```

Step-2: Attempt to ping both devices to verify whether they are responsive or not.

```
(kali㉿kali)-[~]
$ ping 192.168.31.165
PING 192.168.31.165 (192.168.31.165) 56(84) bytes of data:
64 bytes from 192.168.31.165: icmp_seq=1 ttl=128 time=0.879 ms
64 bytes from 192.168.31.165: icmp_seq=2 ttl=128 time=0.984 ms
64 bytes from 192.168.31.165: icmp_seq=3 ttl=128 time=0.695 ms
64 bytes from 192.168.31.165: icmp_seq=4 ttl=128 time=1.20 ms
64 bytes from 192.168.31.165: icmp_seq=5 ttl=128 time=1.01 ms
64 bytes from 192.168.31.165: icmp_seq=6 ttl=128 time=1.09 ms
64 bytes from 192.168.31.165: icmp_seq=7 ttl=128 time=1.12 ms
64 bytes from 192.168.31.165: icmp_seq=8 ttl=128 time=0.676 ms
64 bytes from 192.168.31.165: icmp_seq=9 ttl=128 time=1.05 ms
64 bytes from 192.168.31.165: icmp_seq=10 ttl=128 time=1.14 ms
64 bytes from 192.168.31.165: icmp_seq=11 ttl=128 time=1.02 ms
64 bytes from 192.168.31.165: icmp_seq=12 ttl=128 time=0.891 ms
64 bytes from 192.168.31.165: icmp_seq=13 ttl=128 time=1.17 ms
64 bytes from 192.168.31.165: icmp_seq=14 ttl=128 time=0.692 ms
64 bytes from 192.168.31.165: icmp_seq=15 ttl=128 time=0.838 ms

PS C:\Users\khatr> ping 192.168.236.128

Pinging 192.168.236.128 with 32 bytes of data:
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64
Reply from 192.168.236.128: bytes=32 time<1ms TTL=64

Ping statistics for 192.168.236.128:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
PS C:\Users\khatr> |
```

Step-3 : Please create an iptables rule to stop communication between both devices, as shown in the picture.

```
(kali㉿kali)-[~]  
$ sudo iptables -A INPUT -s 192.168.31.165 -j DROP
```

Step-4 : Now try to ping on both devices its working or not.

```
(kali㉿kali)-[~]  
$ ping 192.168.31.165  
PING 192.168.31.165 (192.168.31.165) 56(84) bytes of data.  
^C  
— 192.168.31.165 ping statistics —  
19 packets transmitted, 0 received, 100% packet loss, time 18439ms
```

C:\Users\Rakshith>ping 192.168.185.141

Pinging 192.168.185.141 with 32 bytes of data:  
Request timed out.  
Request timed out.  
Request timed out.  
Request timed out.

Ping statistics for 192.168.185.141:

Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

Step-5 : Again if you want resume the communication then you have to write another rule to “**ACCEPT**” the communication both device by iptable rule. Before you accept the rule We have to remove the previous rule which we have given the “**DROP**” rule.

```
(kali㉿kali)-[~]  
$ sudo iptables -D INPUT -s 192.168.31.165 -j DROP  
  
(kali㉿kali)-[~]  
$
```

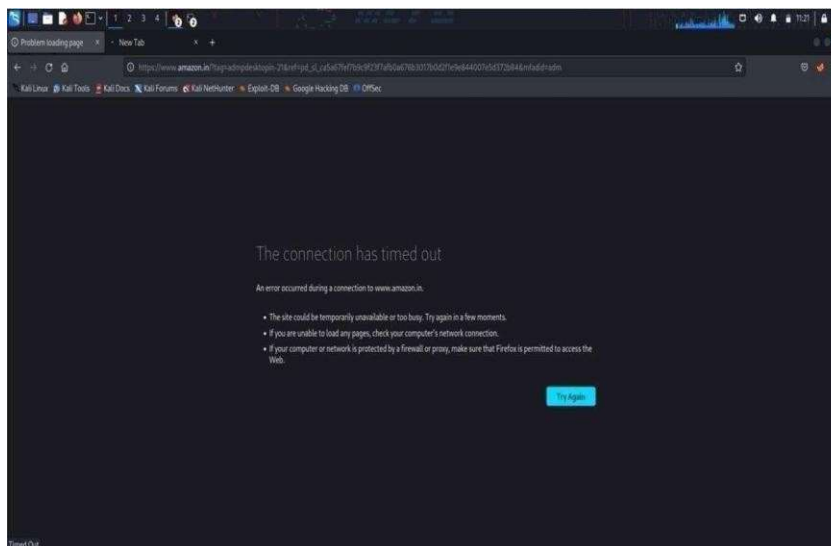
```
(kali㉿kali)-[~]  
$ ping 192.168.31.165  
PING 192.168.31.165 (192.168.31.165) 56(84) bytes of data:  
64 bytes from 192.168.31.165: icmp_seq=1 ttl=128 time=1.26 ms  
64 bytes from 192.168.31.165: icmp_seq=2 ttl=128 time=0.962 ms  
64 bytes from 192.168.31.165: icmp_seq=3 ttl=128 time=1.26 ms  
64 bytes from 192.168.31.165: icmp_seq=4 ttl=128 time=1.31 ms  
64 bytes from 192.168.31.165: icmp_seq=5 ttl=128 time=1.59 ms  
^C  
— 192.168.31.165 ping statistics —  
5 packets transmitted, 5 received, 0% packet loss, time 4007ms  
rtt min/avg/max/mdev = 0.962/1.277/1.591/0.199 ms  
  
(kali㉿kali)-[~]  
$
```

Step-6: To remove the rule from iptable as see in picture.

```
(kali@kali)-[~]  
$ sudo iptables -D INPUT 1  
[sudo] password for kali:  
iptables: No chain/target/match by that name.
```

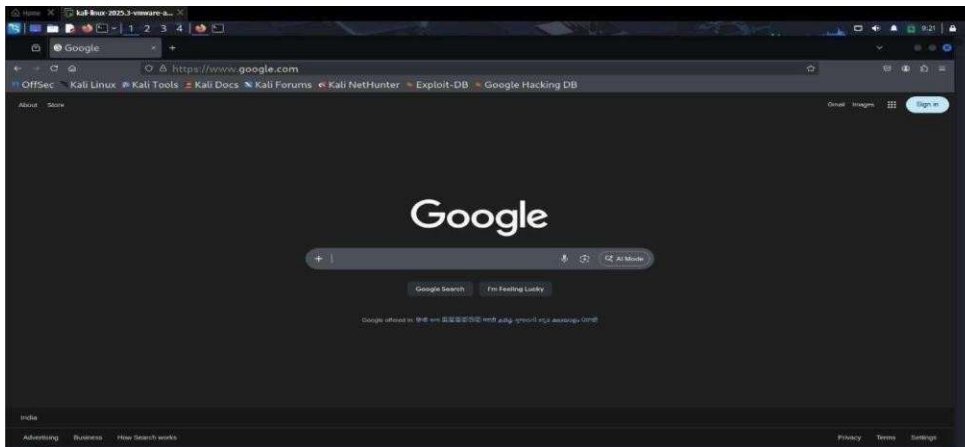
Step-7: To stop https request in our browser by ip address.

```
(kali@kali)-[~]  
$ sudo iptables -A OUTPUT -p tcp -o eth0 -s 192.168.236.128 --dport 443 -j DROP  
(kali@kali)-[~]  
$
```



Step-8 : To resume the https request in browser by accept rule

```
(kali㉿kali)-[~]  
$ sudo iptables -A OUTPUT -p tcp -o eth0 -s 192.168.236.128 --dport 443 -j ACCEPT  
(kali㉿kali)-[~]  
$
```



## **PRACTICAL NO:3**

**Aim :** Implementation of Steganography.

**Scope :** Encompasses the detection of concealed messages and data within various media files, emphasizing the application of steganographic methods for security and forensic purposes.

**Requirements :** The requirements are Hardware and Devices, Steg software, S-Tools, sample image, Documentation Templates etc.

- **Theory:**

### **Steganography**

- The root “steganos” is Greek for “hidden” or “covered,” and the root “graph” is Greek for “to write.” Steganography is the practice of hiding a secret message inside of (or even on top of) something that is not secret.
- Examples of steganography involve embedding a secret piece of text inside of a picture. Or hiding a secret message or script inside of a Word or Excel document. The purpose of steganography is to conceal and deceive. It is a form of covert communication and can involve the use of any medium to hide messages.
- It’s not a form of cryptography, because it doesn’t involve scrambling data or using a key. Instead, it is a form of data hiding and can be executed in clever ways.

### **Types of Steganography?**

#### **1. Text steganography**

Text steganography conceals a secret message inside a piece of text. The simplest version of text steganography might use the first letter in each sentence to form the hidden message. Other text steganography techniques might include adding meaningful typos or encoding information through punctuation.

#### **2. Image steganography**

In image steganography, secret information is encoded within a digital image. This technique relies on the fact that small changes in image color or noise are very difficult to detect with the human eye. For example, one image can be concealed within another by using the least significant bits of each pixel in the image to represent the hidden image instead.



### 3. Video steganography

Video steganography is a more sophisticated version of image steganography that can encode entire videos. Because digital videos are represented as a sequence of consecutive images, each video frame can encode a separate image, hiding a coherent video in plain sight.

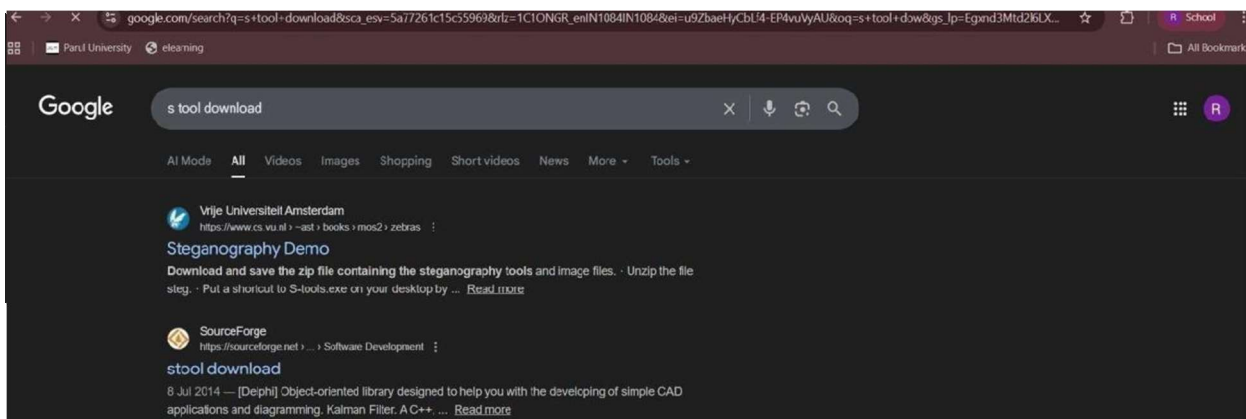
### 4. Audio steganography

Audio files, like images and videos, can be used to conceal information. One simple form of audio steganography is “backmasking,” in which secret messages are played backwards on a track (requiring the listener to play the entire track backwards). More sophisticated techniques might involve the least significant bits of each byte in the audio file, similar to image steganography.

### 5. Network steganography

Last but not least, network steganography is a clever digital steganography technique that hides information inside network traffic. For example, data can be concealed within the TCP/IP headers or payloads of network packets. The sender can even impart information based on the time between sending different packets.

- **Download S tools**





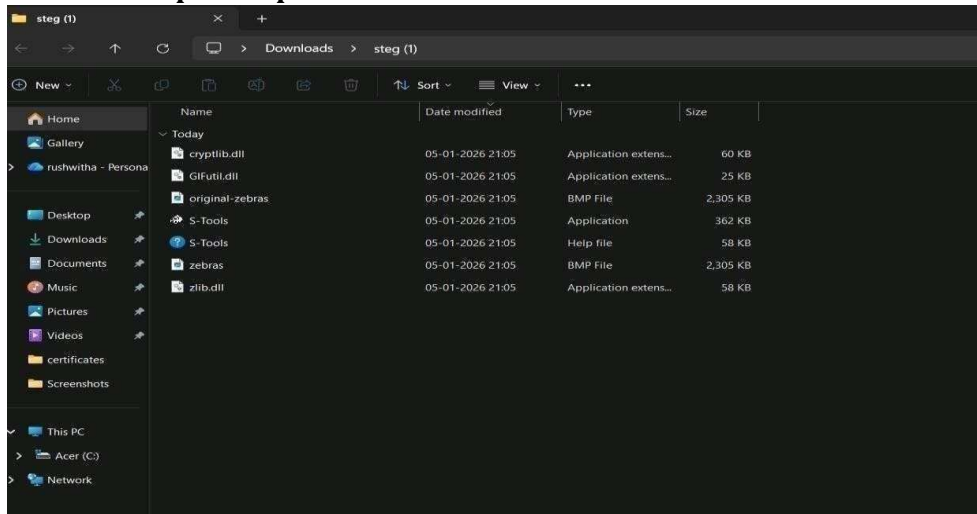
**Steganography Demo**

To try this demo you need a Windows 98/Me/NT/2000/XP/Vista/Win 7 computer. Please follow the instructions below.

1. Download and save the zip file containing the [steganography tools and image files](#).
2. Unzip the file *steg.zip* into an empty directory. If you do not have the *unzip* program already, you can get one [here](#).
3. Put a shortcut to *S-tools.exe* on your desktop by dragging it from Windows Explorer.
4. Drag the *zebras.bmp* file to your desktop. Do not make a shortcut. The file itself must be moved there.
5. Start *S-tools.exe* by double clicking on the icon on the desktop. A window will appear.
6. Drag the *zebras.bmp* file to the S-tools window.
7. Right click on the zebras pictures and select Reveal from the menu.
8. Fill in the 3-character pass phrase 'abc' (without the quotes) in two places. Leave IDEA as the encryption algorithm. Click on OK.
9. Wait until the Revealed Archive dialog box appears. This may take a minute or two.
10. Right click on any item and select Save As to save the file. Repeat for the other ones. These are the hidden files.
11. The file *original-zebras.bmp* is the file before the steganography was done, in case you wish to compare the 'before' and 'after' images.

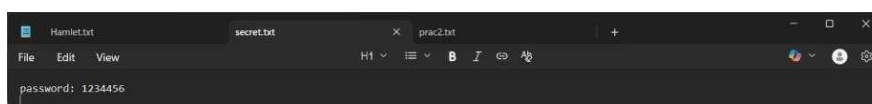
Note: The image at the left has been compressed for quick downloading of this page. It does not contain any secret messages so do not download it unless you really like zebras or [my photography](#).

## Extract the zip file in particular folder



## • Examination:

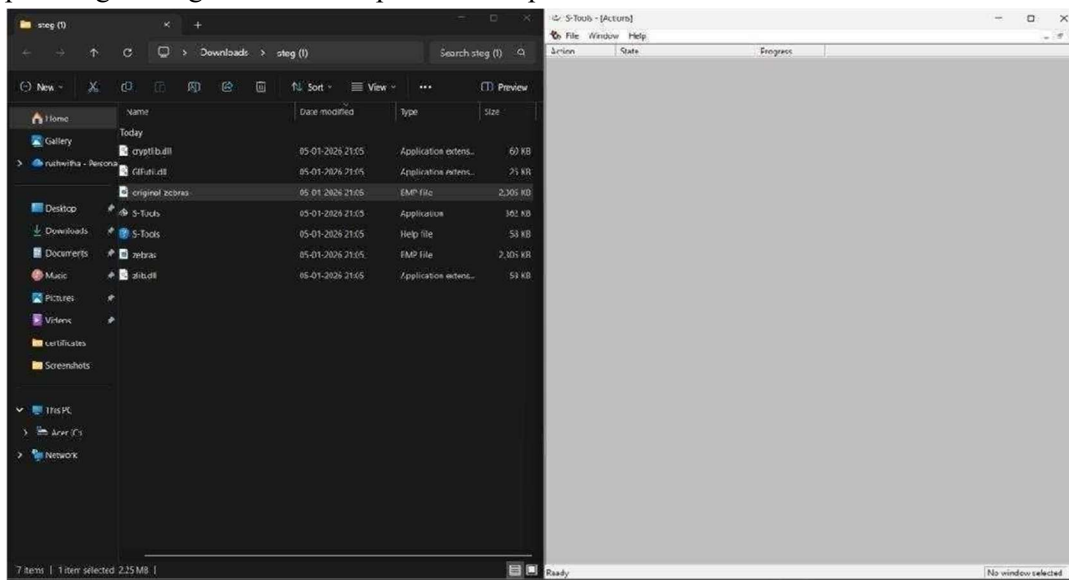
Step-1: Write a secret message in .txt format (Ex: evidence a.txt).



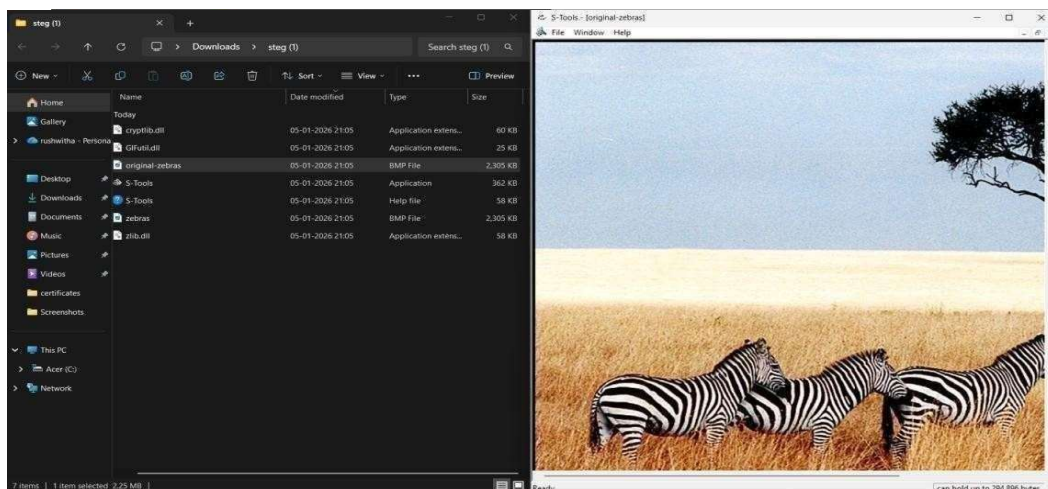
Step 2: Open the S tools



Step 3: Drag the original-zebras.bmp file and drop it into S tools.



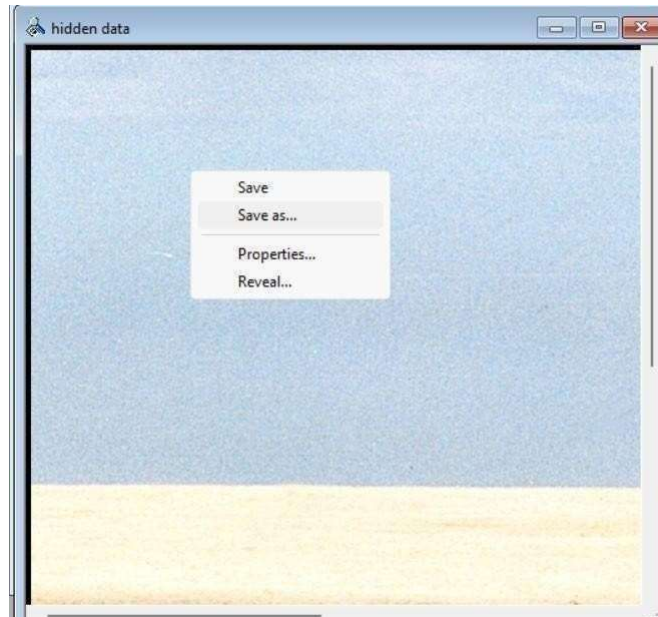
Step 4: Using S-Tools, drag and drop the secret message file on top of image file.



Step 5: Now to encrypt the stego file. Create a new password for the encryption and click on the “OK” button.



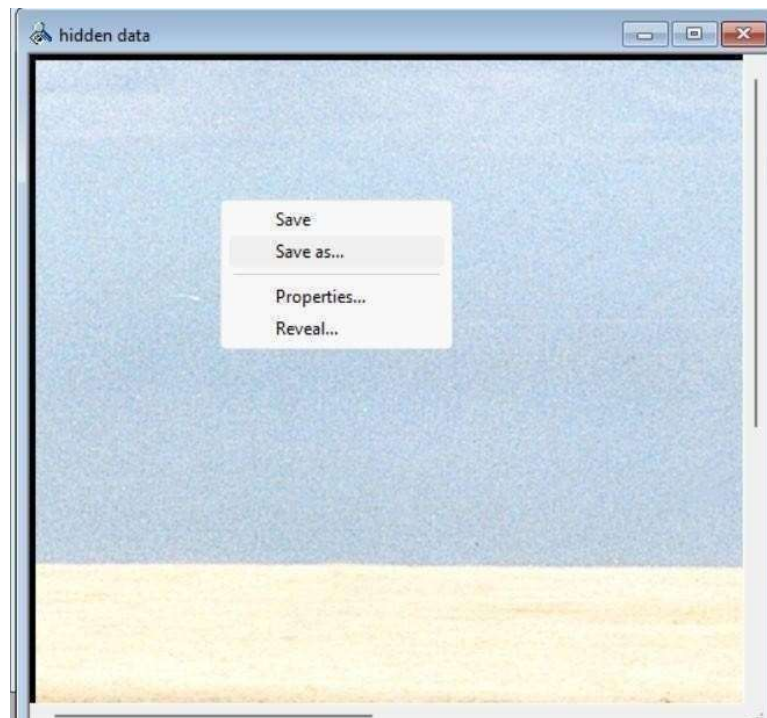
Step 6: generate a new stego image. To save the stego file, right-click on the image and select the “Save as...” option.



Step 7: Specify the destination to save the stego image and click on the “Save” button in order to save the file use .BMP extension.



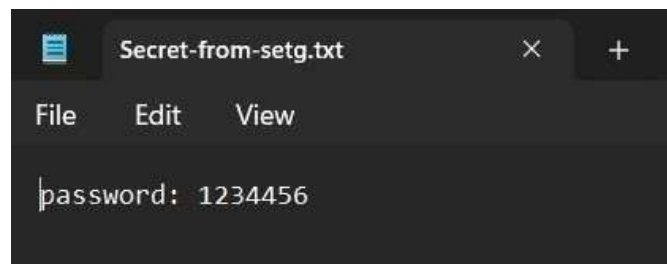
Step 8: Now, to extract the concealed information from the stego image, run the S-Tools and then drag and drop suspected file into it. Right-click on the image and select the “Reveal” option from the top-down menu.



Step 9: “Revealed Archive” window displaying the secret message file name.



Step 10 : Now, Right-click on the file name and then select the “Save as...” option to save it in a location. Next check the extracted file.





## PRACTICAL NO: 4

**Aim: Implementation of MITM- attack using Wireshark / network sniffers.**

**Tools:** Wireshark or Ettercap

We'll need a client machine as well whose network traffic we will spoof and sniff to get cleartext submission of passwords from certain vulnerable websites.

- **Ettercap: Ettercap is GUI based tool built into Kali so need to download and install anything, so let's get started doing a MiTM attack with Ettercap.**

Start Ettercap

Let's view the help file for ettercap by typing; kali > ettercap -h

```

root@kali:~# ettercap -h

ettercap 0.8.0 copyright 2001-2013 Ettercap Development Team

Usage: ettercap [OPTIONS] [TARGET1] [TARGET2]

TARGET is in the format MAC/IP/PORTs (see the man for further detail)

Sniffing and Attack options:
-M, --mitm <METHOD:ARGS>    perform a mitm attack
-o, --only-mitm              don't sniff, only perform the mitm attack
-b, --broadcast              sniff packets destined to broadcast
-B, --bridge <IFACE>        use bridged sniff (needs 2 ifaces)
-p, --nopromisc              do not put the iface in promisc mode
-S, --nossllmitm            do not forge SSL certificates
-u, --unoffensive            do not forward packets
-r, --read <file>            read data from pcapfile <file>
-f, --pcapfilter <string>    set the pcap filter <string>
-R, --reversed               use reversed TARGET matching
-t, --proto <proto>         sniff only this proto (default is all)
--certificate <file>        certificate file to use for SSL MiTM
--private-key <file>        private key file to use for SSL MiTM

User Interface Type:
-T, --text                  use text only GUI
-q, --quiet                 do not display packet contents
-s, --script <CMD>          issue these commands to the GUI
-C, --curses                use curses GUI
-D, --daemon                daemonize ettercap (no GUI)
-G, --gtk                   use GTK+ GUI
  
```

As you can see, ettercap has a significant help file for running it from a command line, but the only thing we need from here is the switch to run it in graphical mode. In the bottom line of the screenshot (not the bottom line of the actual help file as I have truncated it in the interest of space), you can see the -G switch. This after the command ettercap will launch the ettercap GUI.

Start Ettercap GUI

kali > ettercap -G

When we do, the ettercap GUI will start as seen below.



The first step in launching our MiTM attack is to start sniffing. Go to pulldown menu that says "Sniff" and click on "Sniffing at startup".

When we do that, it asking us what interface we want to use and defaults to eth0.

Then click on TICK mark



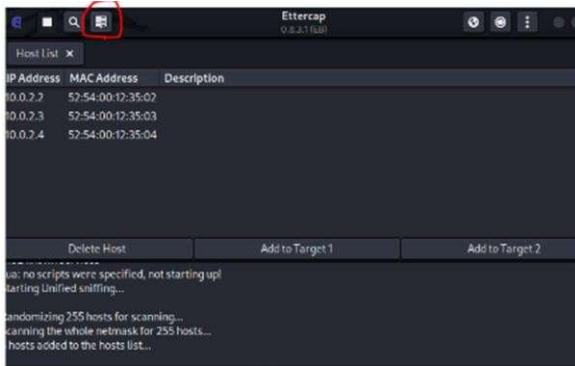
When we click "OK", ettercap launches it sniffing and loads its plugins.

Our next step to find the hosts on the network. Click on the "Hosts" tab and you will see a menu that includes "Scan for Hosts". Click on it and ettercap will begin scanning the network for hosts.



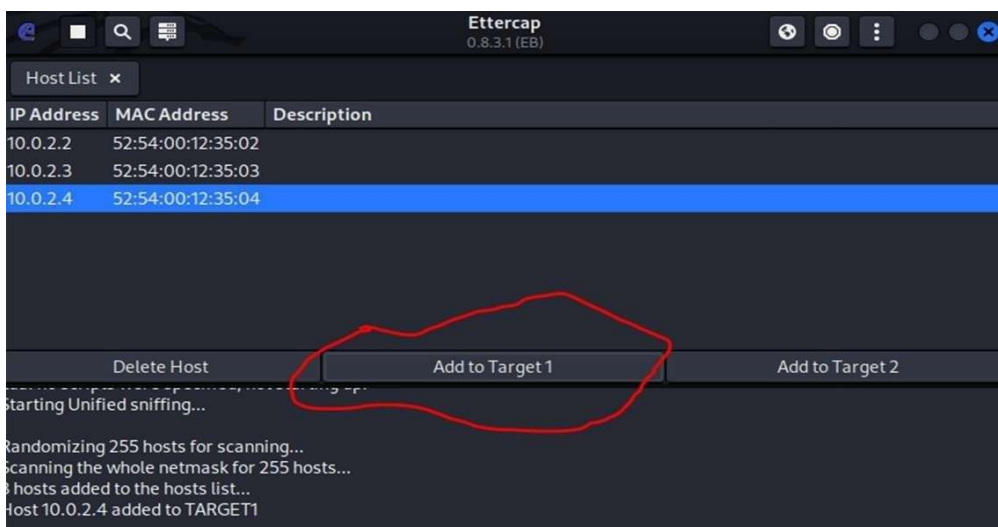
Now, using that same "Hosts" tab, click on "Hosts List". This will display all the hosts that ettercap has discovered on your network as seen in the screenshot below.

En no :2303031250067

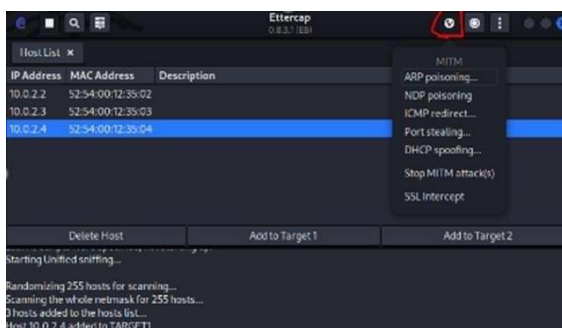


Now, select one of the hosts that will be the target of this attack in the window by clicking on it and then click on "Add to Target 1" at the bottom of the window. When you do so, ettercap will add that host as the first target in our MiTM attack as seen in the screenshot below.

Next, select the second host in this attack and then click "Add to Target 2".



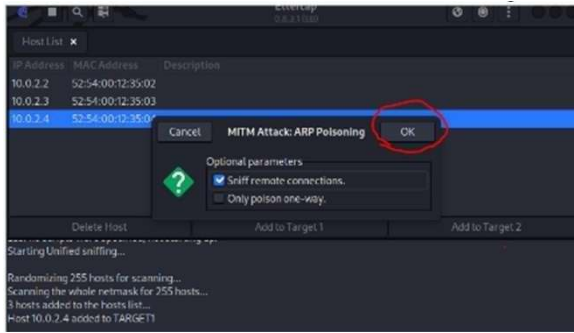
Finally, go to the menu above and click on MITM tab and the drop down menu will have a selection called "ARP Poisoning" as seen in the screenshot below.



Select it and it will open a pop window like below. Select "Sniff remote connections". When we press OK, ettercap will begin ARP poisoning and you will see ettercap respond in its main windows with the message below.

Now, we have successfully placed ourselves between the two targets systems and all their traffic must flow through us. This is where the fun begins as we can now delete, manipulate, impersonate and view





Host 192.168.1.116 added to TARGET1

Host 192.168.1.103 added to TARGET2

ARP poisoning victims:

GROUP 1 : 192.168.1.116 08:00:27:46:69:8D

GROUP 2 : 192.168.1.103 70:1A:04:F4:B9:D0

Now, open any browser on target IP address system and open any HTTP websites of LOGIN PAGE type your id and password then it will show output in ettercap output panel.

## • Wireshark

A basic setup is complete and victim network traffic will now pass through the attacker machine. To listen to these packets, we will use Wireshark.

- Open up a new terminal and type wireshark. Go to the interface which is capturing all the data flow (here eth0) and start the capture.
- Filter out packets according to what you are looking for. For the purpose of this demo, the user is logging in to a vulnerable website DVWA which uses HTTP instead of the secure version HTTPS. Filter protocol as http and search for required data.

**Disclaimer: This tutorial is purely intended for educational purposes and should not be misused.**

- Right click on the packet and follow TCP stream to open up the data contained within. We can clearly obtain the login credentials of the user, that is the username and password.

MITM is one of the classic hacks and on a LAN connection, ARP spoofing is much preferred. Today there have been various measures to prevent such an attack by use of HTTPS, use of VPN and, strong WEP/WAP encryption on access points.

## **PRACTICAL NO: 5**

**Aim :** Implementation of Windows security using firewalls and other tools.

**Tools :** Implementation of Windows security using firewalls and other tools.

**Description :** Windows Firewall is a security feature that helps to protect your device by filtering network traffic that enters and exits your device. This traffic can be filtered based on several criteria, including source and destination IP address, IP protocol, or source and destination port number. Windows Firewall can be configured to block or allow network traffic based on the services and applications that are installed on your device. This allows you to restrict network traffic to only those applications and services that are explicitly allowed to communicate on the network.

The default behavior of Windows Firewall is to:

- block all incoming traffic, unless solicited or matching a rule
- allow all outgoing traffic, unless matching a rule

### **• Firewall Rules**

Firewall rules identify allowed or blocked network traffic, and the conditions for this to happen. The rules offer an extensive selection of conditions to identify traffic, including:

- Application, service or program name
- Source and destination IP addresses
- Can make use dynamic values, like default gateway, DHCP servers, DNS servers and local subnets
- Protocol name or type. For transport layer protocols, TCP and UDP, you can specify ports or port ranges. For custom protocols, you can use a number between 0 and 255 representing the IP protocol
- Interface type
- ICMP/ICMPv6 traffic type and code

### **• Implementation**

**How can you blocked the program?**

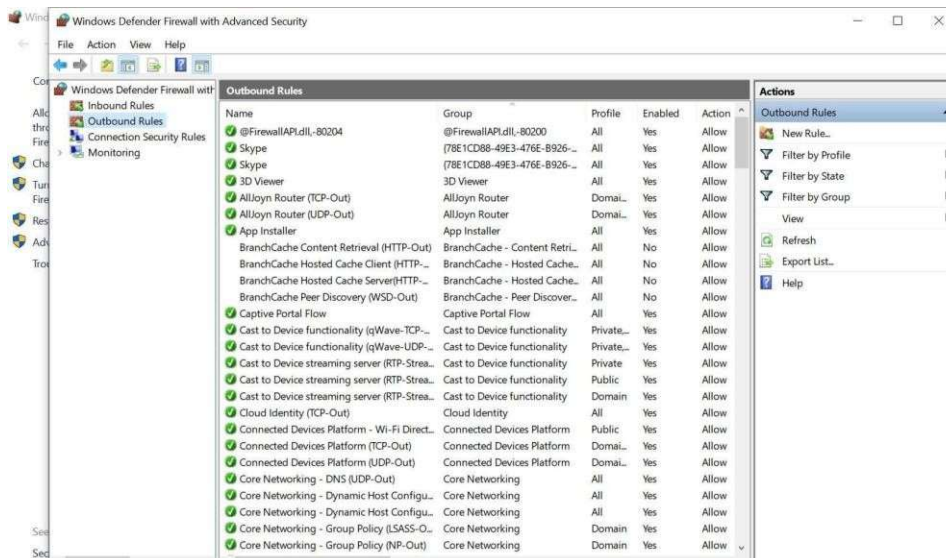
**Step-1 :** Open Windows defender Firewall



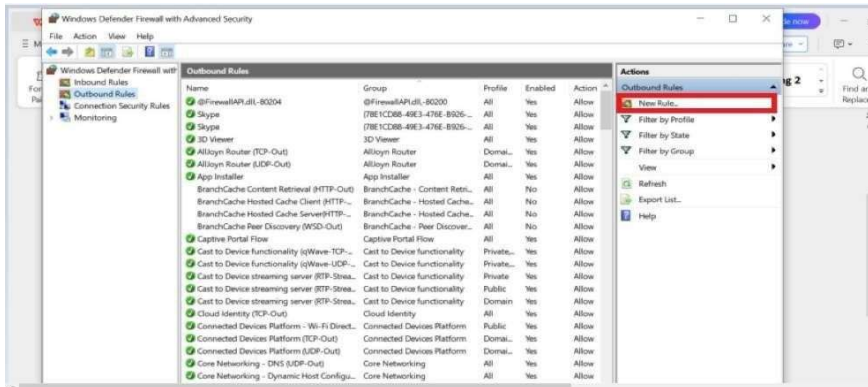
## Step-2 : Select Advance Settings



## Step-3 : Select outbound Rules



## Step-4 : Select New Rule



Step-5 : Select Program and Click on a next Button

